WWII Data Analysis: UPDATE

California CCSS Mathematics Standards Alignment

OBJECTIVES	CODE ASSIGNMENTS	HIGH CONFIDENCE	GRADE BAND
13	**20**	**7**	**9-12**
9 aligned · 4 not	10 distinct standards	10 med · 3 low	inferred from context

A history-context spreadsheet/data-analysis project. Most concrete math is in the function/formula objectives (SUM, MAX, MIN, AVERAGE, XLOOKUP), which map to function-notation standards (F-IF.A.1, F-IF.A.2). The casualty-rate calculation is the strongest ratio/percent task (6.RP/7.RP). Pie- and column-chart objectives lack a clean CCSS-M anchor at HS - MP.4 modeling is the most honest fit. Reflective writing is ELA, not math.

Section 1: V1 Project Setup

Data Test

Project setup verification.

`sections[0].objectives[0]`

Note: Platform setup. No math standard applies.

Platform setup. No math standard applies.

Section 2: Functions

Total Military Personnel

Use SUM() to total the Military Personnel column.

`sections[1].objectives[0]`

F-IF.A.1 F

Grade 9-12

Understand the concept of a function and use function notation.

90 / 100 **HIGH**

Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range. If f is a function and x is an element of its domain, then f(x) denotes the output of f corresponding to the input x. The graph of f is the graph of the equation y = f(x).

WHY THIS MAPS

SUM() maps a domain (range C3:C10) to a single output value - the function-mapping definition.

F-IF.A.2 F

Grade 9-12

Understand the concept of a function and use function notation.

70 / 100 **MEDIUM**

Use function notation, evaluate functions for inputs in their domains, and interpret statements that use function notation in terms of a context.

WHY THIS MAPS

Use function notation and evaluate functions for inputs in their domain - SUM(C3:C10) is function notation in a real context.

A-SSE.A.1.b A

Grade 9-12

Interpret the structure of expressions.

50 / 100 **LOW**

Interpret complicated expressions by viewing one or more of their parts as a single entity. For example, interpret P(1+r) n as the product of P and a factor not depending on P. ■

WHY THIS MAPS

The total can be interpreted as a single quantity that downstream formulas (e.g., casualty rate denominators) operate on.

Total Military Deaths

Copy/paste the SUM formula into the deaths column.

`sections[1].objectives[1]`

F-IF.A.1	F	Grade 9-12	*Understand the concept of a function and use function notation.*	90 / 100 HIGH

Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range. If f is a function and x is an element of its domain, then f(x) denotes the output of f corresponding to the input x. The graph of f is the graph of the equation y = f(x).

WHY THIS MAPS

Same function-mapping concept as Total Military Personnel, applied to a different range.

MP.7	MP	Grade —	*Mathematical Practice*	70 / 100 MEDIUM

Look for and make use of structure.

WHY THIS MAPS

Look for and make use of structure - reusing the SUM-of-range pattern across columns.

Section 3: Formulas

Casualty Rate1

Formula returning Military Deaths / Military Personnel as a percent; fill through E4:E11.

`sections[2].objectives[0]`

7.RP.A.3	RP	Grade 7	*Analyze proportional relationships and use them to solve real-world and mathematical problems.*	**90** / 100 **HIGH**

Use proportional relationships to solve multistep ratio and percent problems. Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.

WHY THIS MAPS

Casualty rate = deaths / personnel * 100% is exactly a percent-of-quantity calculation. Even at HS, this is the underlying ratio-reasoning skill.

6.RP.A.3.c	RP	Grade 6	*Understand ratio concepts and use ratio reasoning to solve problems.*	**70** / 100 **MEDIUM**

Find a percent of a quantity as a rate per 100 (e.g., 30% of a quantity means 30/100 times the quantity); solve problems involving finding the whole, given a part and the percent.

WHY THIS MAPS

Find a percent of a quantity as a rate per 100 - direct match to casualty rate as deaths per 100 personnel.

N-Q.A.1	N	Grade 9-12	*Reason quantitatively and use units to solve problems.*	**70** / 100 **MEDIUM**

Use units as a way to understand problems and to guide the solution of multi-step problems; choose and interpret units consistently in formulas; choose and interpret the scale and the origin in graphs and data displays. ■

WHY THIS MAPS

Choosing units (decimal vs %) and formatting the cell as a percentage.

Section 4: More Functions

Absolute References

Insert MAX() with an absolute row reference over E3:E10.

`sections[3].objectives[0]`

F-IF.A.2 F Grade 9-12 *Understand the concept of a function and use function notation.* **90** / 100 **HIGH**

Use function notation, evaluate functions for inputs in their domains, and interpret statements that use function notation in terms of a context.

WHY THIS MAPS

Use function notation in context - MAX(...) with absolute references is function notation that consistently evaluates over the same input range across copies.

F-IF.A.1 F Grade 9-12 *Understand the concept of a function and use function notation.* **70** / 100 **MEDIUM**

Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range. If f is a function and x is an element of its domain, then f(x) denotes the output of f corresponding to the input x. The graph of f is the graph of the equation y = f(x).

WHY THIS MAPS

MAX is a function from a range to a single value - same domain->range mapping concept.

XLOOKUP Function

Insert XLOOKUP() with relative search_key and absolute lookup/result ranges.

`sections[3].objectives[1]`

F-IF.A.2 F Grade 9-12 *Understand the concept of a function and use function notation.* **90** / 100 **HIGH**

Use function notation, evaluate functions for inputs in their domains, and interpret statements that use function notation in terms of a context.

WHY THIS MAPS

Use function notation, evaluate functions for inputs in their domain - XLOOKUP takes three explicit arguments and produces one output.

F-IF.A.1 F Grade 9-12 *Understand the concept of a function and use function notation.* **70** / 100 **MEDIUM**

Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range. If f is a function and x is an element of its domain, then f(x) denotes the output of f corresponding to the input x. The graph of f is the graph of the equation y = f(x).

WHY THIS MAPS

XLOOKUP defines a mapping from a search key to a result via lookup and result ranges - function-mapping concept in spreadsheet form.

Fill Operation

Use Fill Operation to extend formulas; change D15 to MIN() and D16 to AVERAGE().

`sections[3].objectives[2]`

F-IF.A.2 F Grade 9-12 *Understand the concept of a function and use function notation.* **90** / 100 **HIGH**

Use function notation, evaluate functions for inputs in their domains, and interpret statements that use function notation in terms of a context.

WHY THIS MAPS

Function notation in context - swapping MAX/MIN/AVERAGE over the same input range is exactly evaluating different functions on the same domain.

MP.7 MP Grade — *Mathematical Practice* **70** / 100 **MEDIUM**

Look for and make use of structure.

WHY THIS MAPS

Look for and make use of structure - the Fill Operation propagates the same structural pattern across rows.

Section 5: Applied Knowledge & AI

AI & Critical Thinking

Use MAX/MIN/AVERAGE/XLOOKUP and AI to fill D18:E20.

`sections[4].objectives[0]`

F-IF.A.2 F Grade 9-12 *Understand the concept of a function and use function notation.* **90** / 100 **HIGH**

Use function notation, evaluate functions for inputs in their domains, and interpret statements that use function notation in terms of a context.

WHY THIS MAPS

Composing function-notation calls (MAX/MIN/AVERAGE/XLOOKUP) over absolute ranges to produce correct values is the F-IF.A.2 evaluation skill applied broadly.

MP.5 MP Grade — *Mathematical Practice* **70** / 100 **MEDIUM**

Use appropriate tools strategically.

WHY THIS MAPS

Use appropriate tools strategically - selecting the right function and using AI as a problem-solving tool.

Section 6: Charts

Pie Chart

Create a pie chart of military deaths by country.

`sections[5].objectives[0]`

Note: CCSS-M does not directly anchor pie charts at HS; the 3.MD.C.7.d auto-pick (rectilinear area) is a TF-IDF retrieval artifact, not a real fit.

MP.4 MP Grade — *Mathematical Practice* **70** / 100 **MEDIUM**

Model with mathematics.

WHY THIS MAPS

Model with mathematics - choosing a pie chart to represent the categorical distribution of deaths across countries is a modeling choice.

MP.5 MP Grade — *Mathematical Practice* **50** / 100 **LOW**

Use appropriate tools strategically.

WHY THIS MAPS

Use appropriate tools strategically - selecting and configuring a pie chart for the data.

Column Chart

Create a column chart of military personnel by country.

`sections[5].objectives[1]`

Note: Same retrieval-gap caveat as Pie Chart; 3.MD.C.7.d auto-pick is wrong.

MP.4 MP Grade — *Mathematical Practice* **70** / 100 **MEDIUM**

Model with mathematics.

WHY THIS MAPS

Model with mathematics - a column chart is a modeling choice for representing personnel counts by country.

6.SP.B.4 SP Grade 6 *Summarize and describe distributions.* **50** / 100 **LOW**

Display numerical data in plots on a number line, including dot plots, histograms, and box plots.

WHY THIS MAPS

Display numerical data in plots - middle-grades content standard that arguably extends to column-chart construction in a HS data-analysis context.

Section 7: Reflection

Data Reflection

Use AI to analyze the spreadsheet data and write a 100+ word reflection.

`sections[6].objectives[0]`

Note: Reflective writing in a history/technical context. No math standard applies (closest ELA alignment is WHST.9-10.2/4/8).

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Spreadsheet Reflection

Write a reflection on using spreadsheets in a history class.

`sections[6].objectives[1]`

Note: Reflective writing. No math standard applies.

Reflective writing. No math standard applies.

AI Reflection Write a reflection on using AI in this project. `sections[6].objectives[2]` *Note: Reflective writing. No math standard applies.*	*Reflective writing. No math standard applies.*